Consumer sentiment and economic narratives

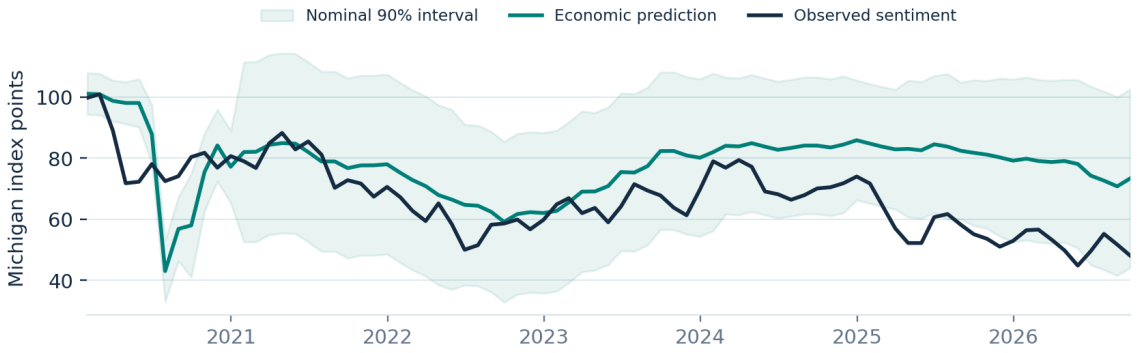
RESEARCH ASSESSMENT / FORECASTING AND RISK

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Economic information cutoff: October 16, 2026. Processing: October 17.

Decision

Keep the original AI tone index in research monitoring. It does not yet show a dependable incremental forecast benefit after household, era, and prior-sentiment information.

Finding on 21 identical test targets	Evidence	Interpretation
A narrow news gain	M1 12.07 / M2 10.82 RMSE	1.25 points lower observed error. The 95% gain interval crosses zero.
Controls change the conclusion	5.82 without tone / 6.06 with tone	The broad model improvement cannot be credited to news.
Recent sentiment is competitive	4.09 last month / 3.98 dynamic controls	Adding AI tone to dynamic controls raises error to 4.32.



Economic prediction and observed sentiment / 81 finalized test months

Research question

Can original AI-measured economic narratives explain or predict weak US consumer sentiment beyond measured economic conditions? The study compares household pressures, media information/framing, and model or survey limitations.

Method in one paragraph

A balanced sample of 670 relevant public titles produces 47 monthly tone readings. Multiple linear regression gives each economic, household, era, and media predictor a separate slope. Timing leaves 45 finalized observations. Expanding fits train on 24 and evaluate the same 21 later targets. A separate long economic exercise evaluates 81 months.

Evidence quality

The computations reproduce. Independent human validation of 300 titles remains unfinished. Historical release vintages are unverified. These gaps prevent a claim of full assignment compliance or production readiness. The results support an exploratory assessment, not a national causal media effect.

01 / Research question and competing explanations

The course brief asks whether measured economic narratives help explain the difference between US sentiment and economic conditions [1]. We test that question without assuming irrational consumers or biased negative reporting.

Required question	Answer supported by this design
1. How can bias be measured?	Compare tone, wording, selection, and numerical claims with a defined event or factual benchmark. Negative tone alone is insufficient.
2. Impact on Michigan sentiment?	Controlled regressions estimate associations. Exposure data are absent, so a national causal impact is unidentified.
3. What do U3, PCE, and GDP explain?	The long-sample model has adjusted R-squared 0.477. Its average gap since 2022 is -14.85 points. Neither number is a causal explained share.
4. Which perceptual mechanisms are measurable?	We measure supplied language and topic emphasis. Reader anchoring, selective exposure, recall, and confirmation bias remain unmeasured.
5. Does media improve prediction?	A small-sample M1 improvement disappears in controlled comparisons. A dictionary sensitivity supplies conflicting exploratory evidence.
6. What contradicts a media-bias explanation?	Accurate favorable news, attenuation with controls, weak changes, and poor incremental forecasts challenge a simple explanation.

Theory and falsification

Household experience predicts that cumulative costs and purchasing power add information beyond annual inflation and GDP. A distinct media signal predicts a contribution after these controls. Model or survey differences predict sensitivity to dynamics, timing, and survey eras.

These explanations overlap. News can transmit genuine household pressures. Controlling for those pressures can absorb part of that channel. Attenuation therefore limits a simple media claim but does not identify the causal alternative.

Earlier versions inspected the evaluation years. The written audit design freezes this correction, but it is not a preregistration or an untouched confirmation. We retain unfavorable results and do not select a method to obtain significance.

02 / Economic inputs and information timing

Input and source	Definition / calculation	Treatment
Michigan headline, supplied workbook	Index of Consumer Sentiment, index points	Outcome. October 2026 preliminary excluded from scoring.
Michigan Conditions / Expectations [2]	Separate published survey components	Secondary outcomes, each with its own prior-month value.
BLS U3, FRED UNRATE [3]	Seasonally adjusted percent of civilian labor force	Monthly source periods kept.
BEA PCEPI [3]	100 x (index / index 12 calendar months earlier - 1)	Headline annual inflation from price levels.
BEA real GDP [3]	Quarter-on-quarter annualized percent change	Quarterly value held after assumed availability. No interpolation.
PCEPI / real disposable income [3]	36-month cumulative price change / annual real-income growth	Separate household controls, not factor scores.
SF Fed news sentiment [4]	Published daily index, averaged over a fixed prior window	External comparator only, not our original NLP measure.

Each prediction uses the first day of its target month. Monthly macro inputs enter after a 45-day buffer from period end. GDP enters after 35 days. The original media index uses the complete month two calendar months earlier, plus seven days for processing.

These buffers avoid obvious period look-ahead but do not recover the values actually released in history. Current GDP, PCE, and income snapshots contain revisions. We call this a retrospective evaluation with assumed availability, not an operational real-time forecast. ALFRED vintage reconstruction remains unfinished [5].

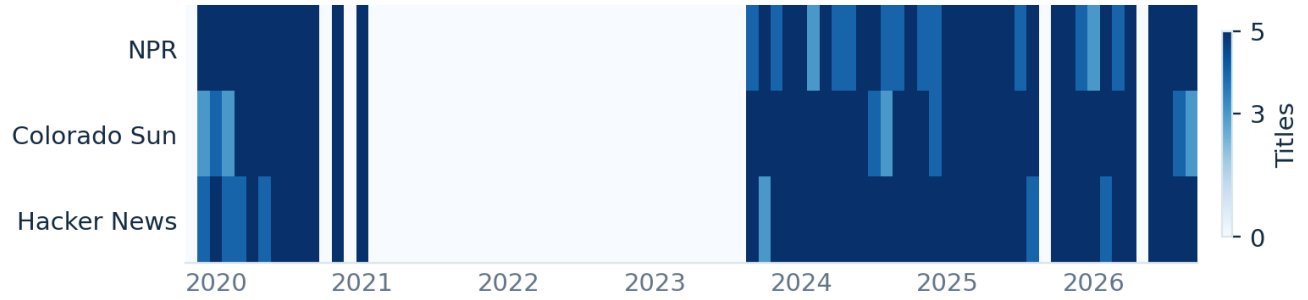
The August 2026 PCE rate in the workbook is 3.00759%. Raw downloaded index levels imply 3.41903%. The cause is unresolved. We consistently use the index-derived definition. Supplied Q3 GDP is excluded because its release falls after the cutoff.

The raw unemployment series has a missing October 2025 value. Selection of the latest nonmissing, assumed-available observation keeps that staleness visible in the input-period columns. Calendar grids remain intact before annual changes. The supplied headline series and calendar lags reproduce exactly.

03 / Original media coverage and sampling

The index uses independently collected public title metadata. No peer data, labels, code, or estimates enter the analysis. The source selection and labels remain independent of peer work.

Stratum / source	Actual retrieval frame	Final titles
National / NPR	Up to 15 visible business-archive cards at four fixed weekly endpoints. This is not a full publication census.	220
State / Colorado Sun	Fixed public Business, Economy, and Housing category union, followed by an economic-title screen.	223
Social / Hacker News	Union of eight fixed monthly searches, followed by economic and US-relevance screens. Submission titles, not user beliefs.	227



Archive access checks covered six outlets. The retained candidate snapshot contains 4,662 rows and 4,604 titles unique within month. Capping at 20 titles per source-month leaves 3,189 candidates. A fixed hash chooses up to five titles from each primary source in balanced windows. A review of 691 selected items excludes 21 irrelevant, foreign-only, or promotional titles.

Each retained source-month has at least three titles. Equal weights within source and then across sources prevent volume shifts between these three outlets from mechanically changing the index. Missing balanced windows remain missing. Gaps are especially important around the inflation surge and constrain generalization.

The unit is the supplied title. The collection preserves metadata excerpts, dates, links, duplicate flags, geography tiers, and failed-window records. One normalized title recurs across different submission months. Exact duplicates are screened within month. Syndication, undetected bots, and forum selection remain possible.

This sample measures selected content supply. It does not measure readership, representative social opinion, or exposure among Michigan respondents. One state outlet and a technology forum cannot represent the whole US media environment.

04 / AI measurement and validation

A written codebook assigns economic direction: -1 for adverse conditions, 0 for mixed or undirected claims, and +1 for favorable conditions. Factual adversity counts as negative tone. Bare policy announcements do not automatically imply a welfare direction. The primary input hides source and date metadata, although titles can reveal context.

Candidate / evaluation	Observed agreement with AI reference	Decision
FinBERT / original 300	47.7% accuracy, macro-F1 0.440	Rejected as primary
Uncalibrated semantic classifier / original 300	55.7% accuracy, macro-F1 0.535	Rejected as primary
Hybrid semantic and conventional text classifier	Grouped development: 68.0%. New 300: 53.3%, macro-F1 0.494	Rejected after transfer failure
Direct assistant coding / 150 repeated items	147 agreements, 98.0%. Kappa 0.968	Provisional measurement, accuracy unverified

The hybrid comparison uses related-story groups during development. The new 300-title transfer reference stays out of its training. The failed transfer is retained, including class/source performance. Both references are AI labels. They do not establish independent human correctness.

The direct code retains all first-pass decisions, 150 second-pass labels, three disagreements, and ten policy clarifications. It reuses 120 exact-record earlier AI labels. Same-session memory and shared errors can inflate agreement. The exact assistant weight snapshot and sampling settings are unavailable. Frozen outputs support numerical reproduction, not identical future AI generation.

Topic attention, price-level/rate framing, emotional words, and political mentions use documented literal proxies. They describe explicit wording, not full semantic meaning. A political mention is not blame. A blame-word co-occurrence is not a validated claim of political bias.

The required independent human validation remains open. Two blinded 300-item files and a sealed source/era mapping are supplied. Humans must label independently, then adjudicate disagreements. The evaluator will calculate agreement and AI class/source performance after actual completion.

Name/source swaps for the rejected hybrid reveal continuous-score sensitivity. A small direct-assistant illustrative review is also retained, but original labels were visible. Neither establishes an unbiased primary measure. These limits restrict all downstream conclusions.

05 / Framing evidence and factual benchmarks

Explicit title vocabulary, %	Colorado Sun	Hacker News	NPR
Prices	17.0	24.2	22.3
Housing	30.9	10.1	13.2
Price-level wording	11.7	1.8	5.0
Inflation-rate wording	2.7	20.3	5.9

The state sample places more explicit emphasis on housing. The forum sample uses more inflation-rate wording. These are shares within selected economic titles. They are not a measure of omitted stories across the whole news agenda, and overlapping vocabulary categories do not sum to 100%.

A benchmark that holds the event fixed

Official event	Captured title evidence	Interpretation
2020 Q2 GDP [6]	NPR: economy drops 32.9%. HN: GDP fell 9.5%.	-32.9% is annualized. The actual quarter change is about -9.49%. Both magnitudes can match the same release.
October 2023 CPI [7]	NPR: inflation cools to 3.2% as gasoline falls.	3.2% is annual, not monthly. BLS monthly seasonally adjusted CPI was 0.0%. Favorable direction can be factually correct.
May 2020 jobs [8]	NPR: 2.5 million jobs added, unemployment dips to 13.3%.	Both numbers match the original release. Employment improvement can coexist with severe remaining hardship.

The GDP conversion is 100 x [(1 - 0.329)^(1/4) - 1] = -9.49%. Omitting annualization in a title creates unit ambiguity. It does not establish a false whole article, because article bodies and reader interpretations are absent.

Four selected title cases are saved with their original URLs and archived official benchmarks. They are purposive examples, not a representative factual-accuracy rate. We do not claim an equivalent-surprise comparison without a surprise dataset.

These cases challenge the assumption that all unfavorable news is biased. They also illustrate why numerical context belongs in the codebook. A credible population estimate of selective coverage requires a much broader event denominator and matched source coverage.

06 / Multiple linear regression and evaluation

Multiple linear regression gives each predictor its own slope. Ordinary least squares minimizes squared errors. The main expanded model has eight predictors and an intercept.

Expanded model M3

$$S(t) = a + b1\ U3 + b2\ PCE12 + b3\ GDP + b4\ Tone(t\text{-}2) + b5\ PCE36 + b6\ Income12 + b7\ Presidency + b8\ WebEra + error$$

S(t) is Michigan sentiment. PCE12 is annual inflation, PCE36 is cumulative prices over 36 months, and Income12 is real-income growth. Economic inputs use their recorded assumed-availability dates. The dynamic variant adds S(t-1).

Specification	Predictors / parameter count including intercept
M1	U3 + annual PCE inflation + quarterly annualized GDP / 4
M2	M1 + original monthly AI tone / 5
Controls without media	M1 + 36-month prices + real-income growth + presidency era + web era / 8
M3	Controls + original AI tone / 9
Dynamic controls / dynamic M3	Corresponding model + prior calendar-month sentiment / 9 or 10
Benchmarks	Last month unchanged. Estimated AR(1) with intercept / 0 fitted or 2 parameters.

The integrated sample contains 45 finalized target observations from February 2020 through September 2026, with gaps. Expanding regressions train initially on 24 observations. All primary models then use the same 21 targets from November 2024 through September 2026. November 2025 and July 2026 lack admissible balanced media.

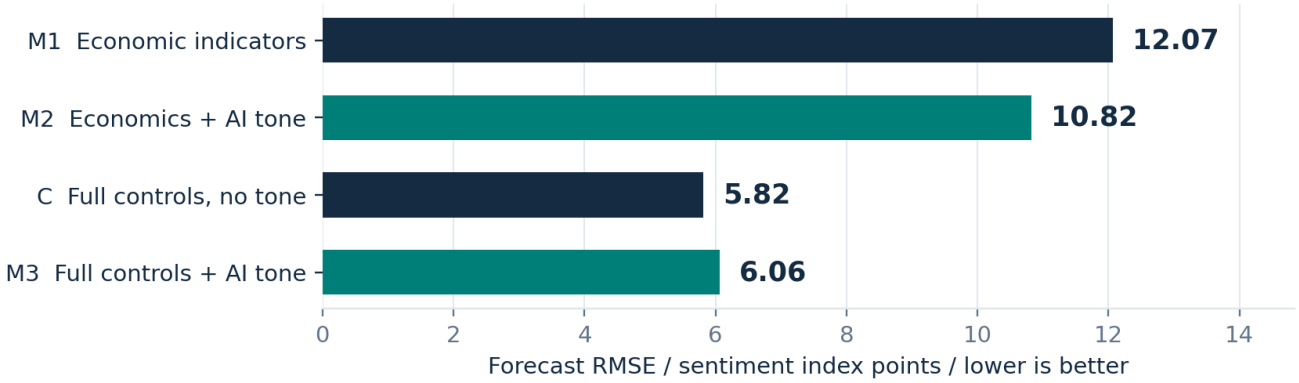
The Democratic-presidency indicator describes an aggregate era. It does not measure respondent party affiliation. The July 2024 web indicator approximates a transition that began in April. Michigan reports that survey-method effects spread over April-July, so a single step is incomplete [2].

A separate long baseline trains on January 2010-December 2019 and evaluates 81 finalized months. Gap = observed sentiment minus the earlier-data economic prediction. The mean gap since January 2022 is -14.85 points. The last 12 finalized months average -25.32 points. Those are model-dependent errors, not measured media effects.

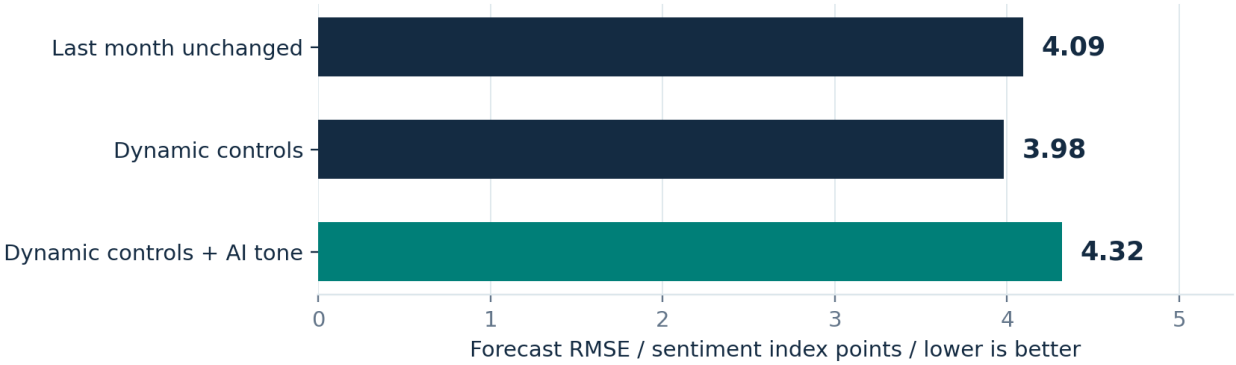
Nominal prediction intervals begin with Student-t OLS uncertainty, then use earlier forecast errors after 12 evaluations. Dependence and changes in the data-generating process can invalidate nominal coverage. The estimated gap also inherits baseline uncertainty. We do not regress it into a causal percentage explained.

07 / Matched forecast comparisons

Lower RMSE means smaller forecast mistakes. All displayed media comparisons use the same 21 finalized target months. Initial training contains 24 observations.



Adding original AI tone reduces the narrow M1 error by 1.25 points, or 10.3%. Its 95% paired block interval is -1.02 to +3.34 points. The improvement is uncertain. M3 reaches 6.06, but identical controls without tone already achieve 5.82.



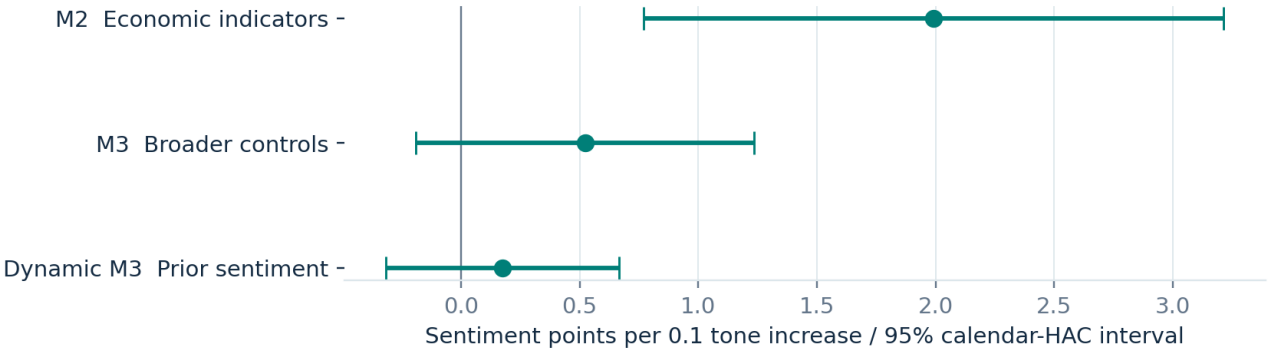
Adding prior sentiment creates a more demanding test. Dynamic controls achieve 3.98, compared with 4.32 after adding tone. Last month unchanged gives 4.09. These are observed comparisons, not proof of a permanent winner.

Separate long economic test / 81 targets	RMSE
Economics only	16.15
Economics + cumulative prices + real income	13.87
Last month unchanged	4.94

The 81-month exercise uses different dates and must not be compared directly with the 21-target media scores. Most media-test evidence concerns 2025-2026. Two missing target months remain excluded. October 2026 is preliminary and unscored.

08 / Coefficients and model risk

These are conditional associations on the 45 finalized media-aligned observations. HAC means heteroskedasticity and autocorrelation consistent uncertainty. The covariance calculation uses actual calendar distances, so missing months do not become adjacent observations.



The narrow association shrinks with plausible alternative explanations and persistence. A 0.1 tone change is a tenth of the -1 to +1 title index. It is not a 10% change in consumer mood or an intervention on exposure.

M3 direct predictor	Slope	95% HAC interval
Unemployment, 1 pp	-3.89	-6.78 to -1.00
Annual inflation, 1 pp	-12.89	-18.95 to -6.84
GDP annualized growth, 1 pp	-0.02	-0.10 to +0.06
36-month prices, 1 pp	-0.50	-1.61 to +0.61
Real-income growth, 1 pp	+1.81	-0.16 to +3.77
Democratic-presidency era	+9.45	+5.44 to +13.46
Web-survey era	-6.05	-10.97 to -1.13

M3 adjusted R-squared is 0.824, but fit does not establish future usefulness. Its residual lag-one calendar correlation is 0.378. Dynamics reduce that correlation to 0.116. Calendar first differences leave 39 observations and weaken the tone relationship. Twelve-month changes leave only 22 and are particularly fragile.

VIF, a measure of predictor overlap, is 16.61 for U3, 14.59 for real income, and 9.62 for cumulative prices. Individual economic and era slopes are unstable or confounded. They cannot be interpreted as causal shares. Deleting one month moves the M3 media slope from 2.59 to 6.37 per full index unit. The highest Cook influence is February 2020.

Nine parameters in M3 and ten in the dynamic model use only 24 initial observations. This is a substantial estimation burden. We report it rather than treat high fit as evidence that the design is strong.

09 / Robustness and contrary evidence

Dynamic comparison	Without tone	With measure	n
Primary direct AI	3.98	4.32	21
Simple dictionary	3.98	3.55	21
Raw semantic model	3.98	4.64	21
Without NPR	3.98	4.21	21
Without Colorado Sun	3.98	4.14	21
Without Hacker News	3.98	4.11	21
Annual seasonal terms	5.33	5.77	21
Three-month media lag	4.60	4.68	20

The dictionary improvement is contrary evidence that matters. We retain it as a future candidate rather than replace the primary index after seeing the result. Its agreement with AI reference labels is weak. A favorable predictive point estimate does not establish that it measures the intended construct.

Current Conditions RMSE rises from 4.74 to 5.02 when primary tone enters the dynamic model. Expectations rises from 4.92 to 5.25. These secondary outcomes use the same 21 targets. The strict-US title subset has too little balanced coverage for the minimum forecast comparison. We do not fill those missing months.

The same-month tone-gap correlation is 0.140 on 46 matched months. Lagged and future-tone diagnostics are small in this dataset, but they do not establish direction of causation. The gap is an estimated outcome, and omitted economic shocks can influence both series.

Forecast uncertainty

Six fixed paired comparisons use 2,000 circular calendar-block bootstrap draws with six-month blocks. Approximate HAC(3) squared-loss tests receive Holm adjustment for multiple testing. The M1-M2 adjusted p-value is 1.000. None of the six comparisons clears 5% after adjustment.

The dynamic-media bootstrap gain interval is negative, while the adjusted loss test remains uncertain. Different estimands and fragile small-sample approximations can disagree. We do not convert either result into proof of harm or zero effect.

The practical contradiction is narrower: this particular AI index does not supply a stable additional forecast benefit across the controlled comparisons. The short sample, selected sources, revised inputs, and unverified labels prevent a general conclusion about all media.

10 / Causal interpretation and next evidence

Elevated prices can depress purchasing power even when inflation falls. A title that emphasizes that hardship can be accurate. Text measures price-level wording, but it cannot distinguish rational household experience from irrational anchoring without corresponding belief data.

Similarly, available negative stories do not prove availability bias in a reader. Confirmation bias requires evidence about prior beliefs and selective consumption. Our title data supply neither. Published Michigan group tables motivate heterogeneity checks, but do not link those groups to our sources.

Michigan reports partisan differences and differences between phone and web surveys [2]. We retain aggregate income, age, and party source tables for descriptive work. We do not estimate individual moderators from national monthly data. Our era indicators absorb many coincident changes and cannot isolate political persuasion or a survey-method effect.

Proposed truthful-framing experiment

After required institutional review, recruit 600 consenting US adults. Randomize within income, baseline sentiment, and party strata. Assign 200 participants to price-level emphasis, inflation-rate emphasis, or balanced emphasis. Every summary contains the same verified values, dates, definitions, and source label from one official release.

Measure baseline household strain and beliefs before treatment. Record immediate economic expectations, perceived purchasing power, factual recall, and a sentiment-related response scale after treatment. Use the level-versus-rate contrast as the primary test. Estimate intention-to-treat effects with baseline adjustment and report confidence intervals.

Prespecify recruitment, exclusions, missing-data handling, attrition, secondary tests, and a seven-day retention check. Correct secondary comparisons. With 200 per arm, an independent-outcome normal approximation gives 80% power near 0.28 standard deviations at a two-sided 5% level. This is a planning calculation, not an observed effect.

Randomization addresses selective exposure within the experiment. Demand effects, noncompliance, spillovers, and artificial exposure remain risks. Moving from a vignette response to the national Michigan index requires representative sampling, exposure frequency, weighting, and transport assumptions.

The present study follows the permitted observational-analysis-plus-causal-proposal route. No experiment ran. Economic shocks, audience demand, and reverse selection prevent a causal interpretation of the current regressions.

11 / Monitoring, recommendation, and sources

The offline dashboard displays sentiment, predictions and nominal intervals, gaps, monthly tone and conditional sampling intervals, literal framing/salience, event records, source counts, and accuracy. It preserves missing values and marks October as preliminary. Users can import/export the documented JSON format. Rerunning the Python analysis regenerates the dashboard.

Future updates must freeze the forecast before seeing its outcome. Append timestamped releases and metadata. Retain old vintages and failures. Finalize labels before refitting. Record errors before changing specifications. A comparison around election day alone does not identify an election effect.

Financial implication

Recent sentiment provides a demanding baseline for forecasts of consumer demand. Household pressures and sentiment gaps can guide scenario review. A text signal needs prospective gains and validated measurement before it supports a forecasting or risk decision.

Final judgment

The narrow macro model is incomplete. Our AI tone measure does not reliably improve controlled forecasts, and the evidence does not establish media bias as the main cause of pessimism. The data also cannot determine the causal shares of household strain, politics, media, and survey changes. Independent human validation and historical-vintage reconstruction remain unfinished.

Sources and reproducibility

[1] Course brief: Final_Project_Applications_in_Finance_Oct_2026.pdf and supplied Perceptual_Bias_Data.xlsx. The package includes a requirement-by-requirement audit.

[2] University of Michigan, Surveys of Consumers. Historical components and group tables: data.sca.isr.umich.edu. Method transition: sca.isr.umich.edu/files/methodtransitionannouncement2024.pdf (April 5, 2024).

[3] BLS/BEA official data via FRED: fred.stlouisfed.org/series/UNRATE, PCEPI, DSPIC96, A191RL1Q225SBEA. Frozen raw files and hashes accompany the analysis.

[4] San Francisco Fed, Daily News Sentiment Index: frbsf.org/research-and-insights/data-and-indicators/daily-news-sentiment-index/. External benchmark.

[5] FRED real-time periods: fred.stlouisfed.org/docs/api/fred/realtime_period.html. Historical vintages are not reconstructed in this study.

[6] BEA, July 30, 2020 GDP advance release: bea.gov/news/2020/gross-domestic-product-2nd-quarter-2020-advance-estimate-and-annual-update.

[7] BLS, November 14, 2023 CPI release: bls.gov/news.release/archives/cpi_11142023.htm. [8] BLS, June 5, 2020 employment release: bls.gov/news.release/archives/empsit_06052020.htm.

[9] Original metadata routes: npr.org/sections/business/archive, coloradosun.com/wp-json/wp/v2/posts, hn.algolia.com/api/v1/search_by_date. Queries, cached responses, and access failures are retained.

[10] Rejected classifier manifests: Xenova/finbert (ProsusAI base) and Xenova/nli-deberta-v3-small. Model revisions, prompts, labels, and failures remain in the audit folder. Final direct-AI snapshot is unavailable.